Timothy Jin

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EDUCATION

University of British Columbia | *B.Sc. Computer Science — 3.9 / 4.0 GPA* — Sep 2021 – May 2026

Data Structures & Algorithms · Machine Learning · Networking & Security · Databases · Distributed Systems

EXPERIENCE

Amazon | *Software Development Engineer Intern* — May 2025 – Aug 2025

- **Eliminated engineer intervention for 80% of support tickets** submitted by internal teams for help with the Prime Subscriptions API — built an agentic assistant in **Java, Python, and TypeScript** powered by **AWS and Claude LLMs** to automatically analyze issues, surface insights, and run API tests.
- **Reduced debugging time by 80%** by migrating the Prime Subscriptions API from compressed file-based logs (requiring manual grepping) to centralized, queryable **CloudWatch logging** across **9 global deployments**.
- **Reduced manual scaling work by 95%** for high-traffic shopping events (e.g. Prime Day) by integrating the Prime API into an automated capacity planning interface, replacing a process that previously required weeks of manual effort.

Stanford University — Dept. of Emergency Medicine | *Machine Learning Researcher* — May 2024 – Oct 2025

- Improved emergency department patient outcome prediction by **0.12 AUROC** by tuning and deploying **LightGBM** models on historical patient data, enabling better resource allocation for **100+ patients daily**.
- Identified the optimal model architecture — **15% better accuracy** — by benchmarking PyTorch transformer and LLM-assisted LightGBM models against baselines across **millions of patient records**.
- Improved ED physician confidence in model decisions by building a **React + AWS dashboard** with **SHAP-based explainability**, making predictions transparent and actionable for clinical teams.

Rivian | *Software Engineer Intern* — May 2024 – Dec 2024

- Maintained **100% feature test coverage** during Rivian's migration from a legacy messaging system to a faster **Protocol Buffers** framework — built Rivian's first **Python Protocol Buffers pub/sub framework** and NATS client tool, enabling hundreds of engineers to test against virtual cloud-based vehicles without relying on scarce physical hardware.
- Proposed and built an **agentic full-stack vehicle simulator** (React, TypeScript, AWS, Claude LLM) that let **100+ engineers, designers, and managers** test mobile and cloud features without hours of authentication and CLI setup.
- Led mobile integration of the 2024 Halloween feature and multiple other releases, shipping to **100,000+ customers** across **7 major app versions** by owning test strategy and driving cross-team alignment.

UBC Cloud Innovation Centre (AWS) | *Software Developer Co-op* — Jan 2023 – Aug 2023

- Built a **Flutter/Dart mobile app** with an **AWS TypeScript** serverless backend to collect Parkinson's disease symptom data from **thousands of patients annually** with minimal discomfort, supporting clinical research teams.
- Improved balance assessment for rehabilitation patients using **transformer-based ML** on IMU sensor data, delivered via a **React and Flutter UI** with a TypeScript/Python AWS backend.
- Presented at the **2024 Interdisciplinary Conference in Rehabilitation Science BC** (INCREASE BC).

PROJECTS

FinHog · *Agentic AI, TypeScript, Python, AWS, React* — Sep 2025

- Won **1st place at HackMIT** (1,500 participants) for Anthropic's Best Use of Claude, and **2nd place** for Warp's (YC W23) Best Financial Visualization Agent.
- Built an **autonomous financial analytics agent** that analyzes data in real time and automatically selects and renders the most informative visualizations — delivered in under 24 hours.

Vancouver Transit App · *Dart, Flutter, iOS, Android* — Nov 2019 – Jan 2022

- Shipped an iOS and Android app providing real-time transit data, reaching **1,000+ active users** and **10,000+ installs** across the Apple App Store and Google Play Store.

TECHNICAL SKILLS

Languages	TypeScript, Python, C++, Java, SQL, JavaScript, Dart, GraphQL, HTML/CSS
Frameworks & Libraries	React, Node.js, Flutter, AWS CDK, Pandas, NumPy, Protocol Buffers, JUnit
Tools & Platforms	AWS, PostgreSQL, MySQL, Docker, Git, CI/CD, NoSQL, Xcode, Android Studio